REPORT FRONT SHEET

MOLECULAR BIOLOGY, Master Course
EXPERIMENTAL CELL BIOLOGY

NAME OF EXPERIMENT: Exercise 2 A/B week 1-2: Cell Cycle Analysis and Hoechst 33342 / Alamar Blue 96 well assays

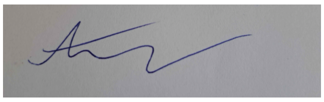
DATE FOR PERFORMANCE OF EXPERIMENT: 15-06-2026 -- 26-06-2026

TEACHER: Camilla Michelle Thorup Andersen

TEAM Green GROUP NUMBER: 3

Names and signatures

Andreas Giovanni Kristian Frilev Olesen,



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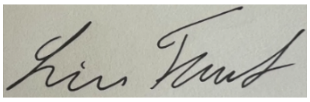

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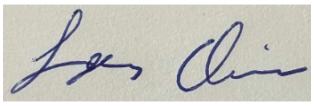


DATE(S) FOR HANDING IN THE REPORT TO THE TEACHER:

Date for 1. Time	Date for 2. time	Date for 3. Time
26.06.2026	29.06.2026	

TEACHER COMMENTS:

REPORT APPROVED BY TEACHER:

Signature Date

This is the front page for the reports and has to be attached to the report. The original front page should be attached when a revised report is handed in.
THE BIOLOGY STUDY BOARD

Contents

Cell cycle analysis (2a):

Introduction

In this experiment, flow cytometry-based cell cycle analysis were used to investigate the effects of sodium butyrate (NaBu) on DLD-1 cell lines, for week 2 resveratrol was also used as treatment. Flow cytometry with propidium iodide (PI) staining allows quantification of cellular DNA content and determination of the distribution of cells in the G1, S, and G2/M phases of the cell cycle. Changes in cell cycle distribution can indicate altered proliferation or cell cycle arrest induced by experimental treatments.

Methods

DLD-1 were seeded in T75 flasks and cultured for 24 h before treatment. Cells were exposed to either PBS control, 10 mM NaBu or 20 μm resveratrol for 48 hours..

Following treatment, cells were harvested by trypsinization, washed in PBS w/o + 5 mM EDTA+ 0.5% BSA (PBS-EB), and fixed in 70% ethanol. Fixed cells were stained with propidium iodide and RNase A before analysis by flow cytometry. Data acquisition and analysis were performed using the free software Floreada.io with cell cycle analysis based on the Watson pragmatic model. Single-cell population was gating on FSC-A versus FSC-H plots. Cell cycle distributions was analysed based on DNA content histograms.

Results week 1

Single-cell populations was gated from FSC-A versus FSC-H. The same gate was applied to all samples to minimize analysis bias, gating is seen in figure 1 for PBS-treatment and figure 2 for NaBu-treatment.

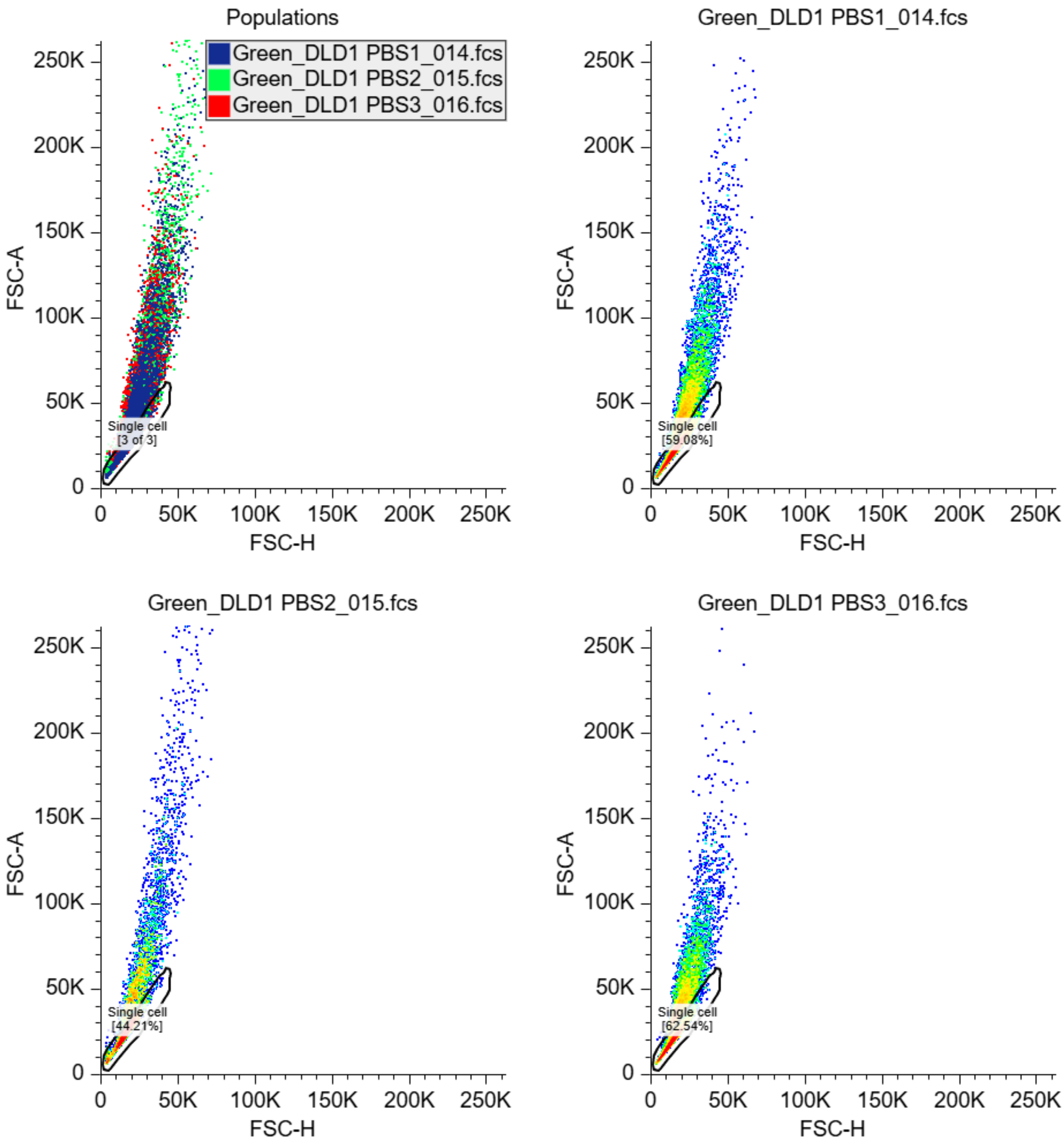


Figure 1. FSC-A plotted against FSC-H for all samples of PBS, and each sample individually. The same gating for single cell, were used for all replicates, to minimize bias. Eventcounts for PBS1= 10.000, PBS2= 3739 PBS2: 10.000 before gating.
After gating is applied PBS1= 5908, PBS2= 1653, PBS3= 6254.
Data acquisition and analysis were performed using the free software Floreada.io.

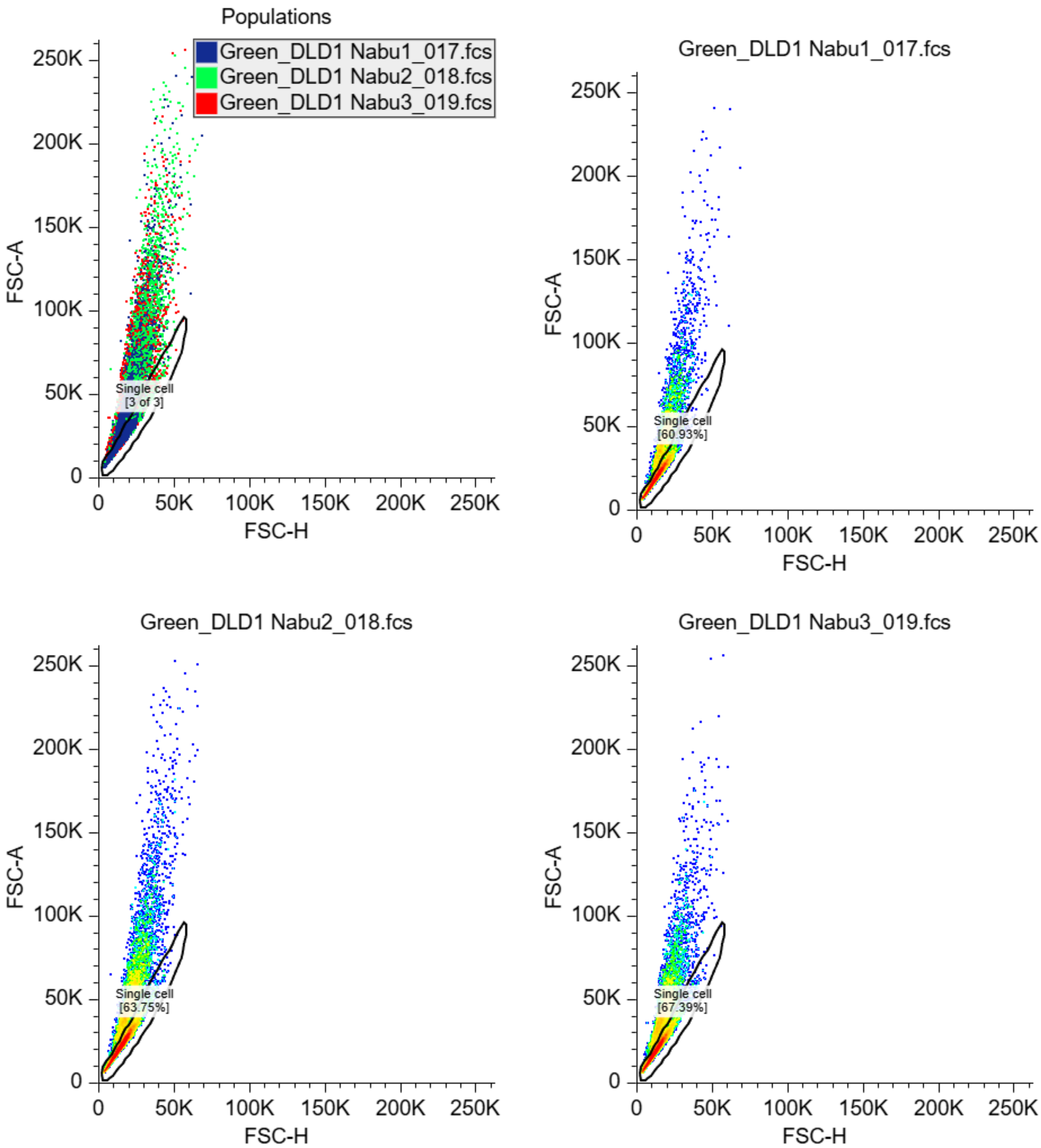
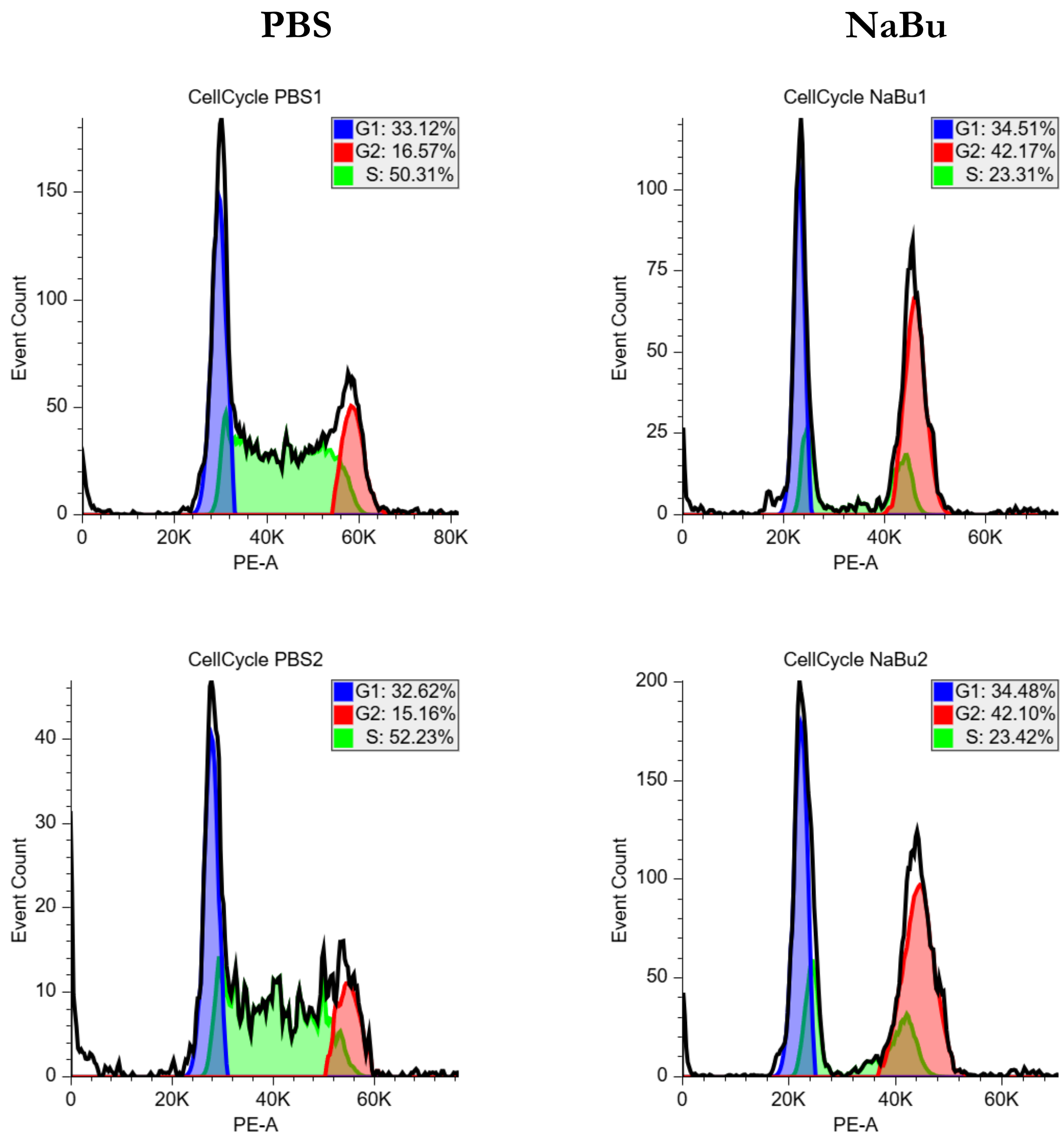


Figure 2. FSC-A plotted against FSC-H for all samples of NaBu, and each sample individually. The same gating for single cell, were used for all replicates, to minimize bias. Event counts for NaBu1= 5076, NaBu2: 10.000, NaBu3=10.000 before gating.
After gating is applied NaBu1= 3093, NaBu2= 6375, NaBu3= 6739
Data acquisition and analysis were performed using the free software Floreada.io.

As observed in figure 3, the PBS-treated control cells showed a relatively similar cell cycle distribution across all three replicate. Approximately 33% of cells were present in G1 phase, 50–52% in S phase, and 15–18% in G2 phase. Treatment with NaBu altered DNA content histograms compared with PBS-treated controls as displayed on figure 3. In the NaBu-treated samples, the population in G2/M phase appeared increased relative to the controls. However, the automatic cell-cycle modelling performed by floreada.io did not accurately fit the NaBu histograms, as the fitted S-phase distribution overlapped substantially with the G2/M. The calculated percentage should be regarded as approximate estimates rather than quantitative measurements. However the visual shift of the histograms supports an accumulation of cells in G2/M following NaBu treatment.



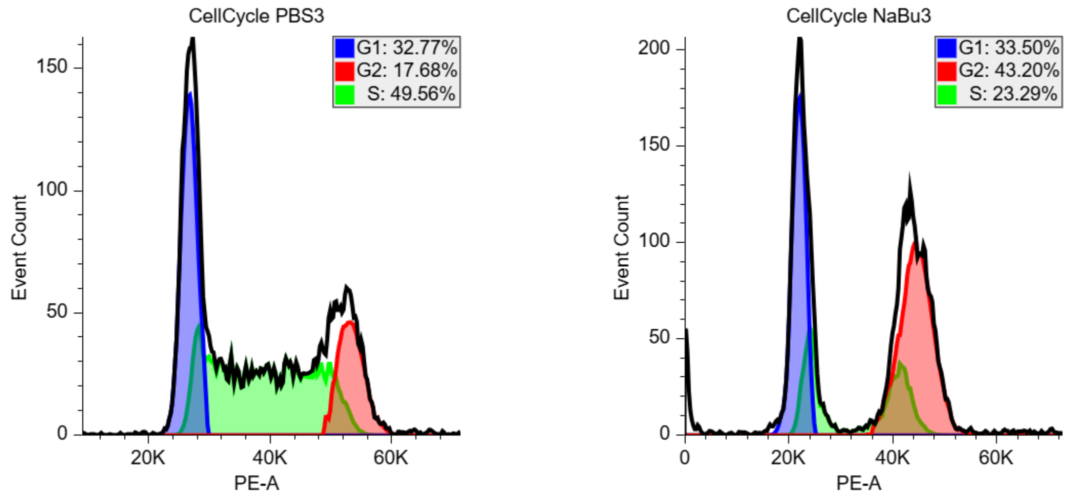


Figure 3. DNA-content histograms of gated single DLD-1 cells after treatment with PBS or 10 mM sodium butyrate (NaBu) for 48 h. Histograms were generated in Floreada.io using the Watson pragmatic model. Although the software estimated the proportion of cells in G1, S and G2/M phases, the automatic fitting did not adequately resolve the histograms because the fitted S-phase distribution overlapped the G2/M peak. Therefore, the displayed percentages should be interpreted as approximate.

	G1	G2	S
PBS1	33,12%	16,57%	50,31%
PBS2	32,62%	15,16%	52,23%
PBS3	32,77%	17,68%	49,56%
	G1	G2	S
NaBu1	34,51%	42,17%	23,31%
NaBu2	34,48%	42,10%	23,42%
NaBu3	33,50%	43,20%	23,29%

Table 1. Estimated cell-cycle phase distribution generated automatically by Floreada.io using the Watson Pragmatic model. Due to imperfect fitting of the Histograms, particularly the overlap between the S-phase and the G2/M-phase, these values should be interpreted only as approximates estimates, rather than precise measurements.

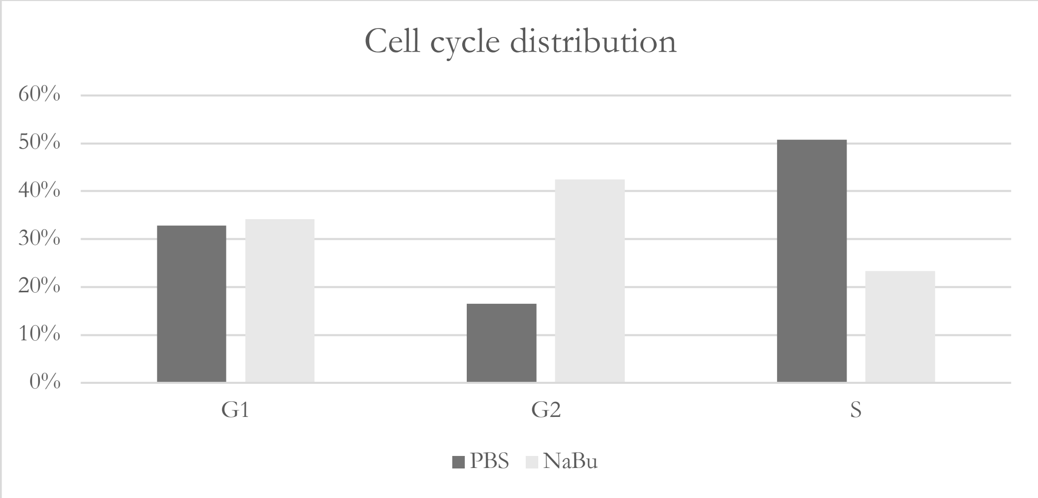


Figure 4. Estimated cell-cycle phase distribution of DLD-1 cells treated with PBS or 10 mM NaBu. Bars represent the mean estimated distribution obtained from Floreada.io. Values should be interpreted qualitatively rather than as exact percentages.

Results week 2

Cell cycle analysis was repeated in week 2 using PBS-treated control cells together with cells treated with 10 mM sodium butyrate (NaBu) or 40 µM resveratrol as displayed on figure 5.

The PBS-treated cells displayed an unusual DNA-content histogram that while retaining the usual G1, S1 and G2 populations showed a strong enrichment in G2 (~40%) when compared to the the data in week 1 (~16%). NaBu treatment visibly altered the DNA-content distribution by increasing the DNA content, but in this case we saw an increase in cells in the S phase compared to week 1. Finally, the Resveratrol treated flask displays a very unusual distribution with almost no cells in the G2 phase and approximately 60% in G1 and 40% in S, consistent with prior knowledge that Resveratrol seems to arrest cells in the S phase. Comparing the distributions with the proportions from group Blue which used Caco-2 (figure 6), we see that Caco-2 cells remain significantly more stable in terms of distributions and don't show the same abrupt shifts that our DLD-1 cells showed, although we do see somewhat of an expansion in the proportion of cells in the S-phase for the samples treated with Resveratrol.

As in the week 1 analysis, the automatic Watson model implemented in Floreada.io did not accurately resolve the histograms. Therefore, the numerical phase percentages were not considered reliable and were not used for quantitative interpretation. Instead, conclusions were based on the qualitative changes observed in the DNA-content histograms.

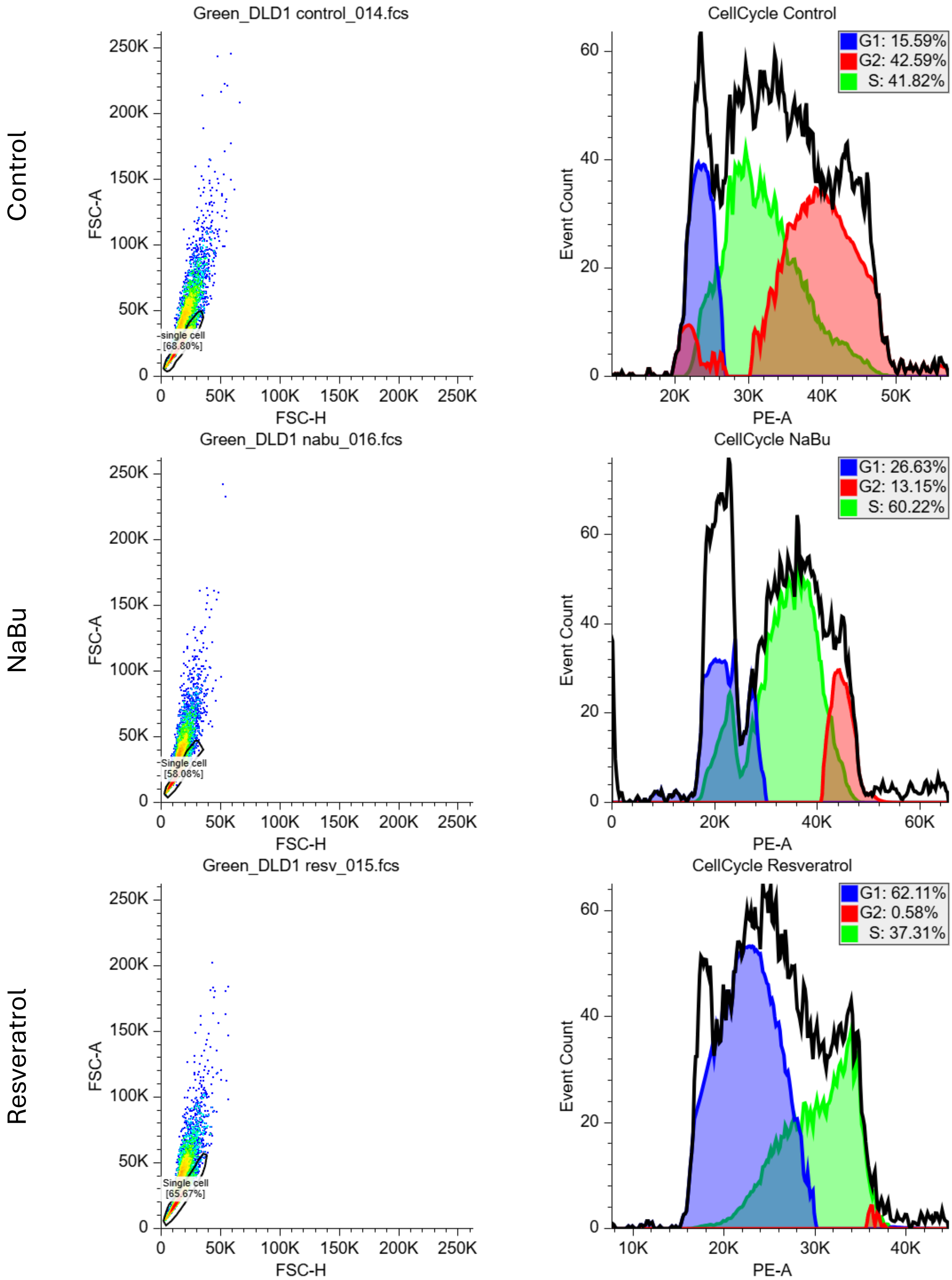


Figure 5. FSC-A plotted against FSC-H and Cell cycle analysis for each sample in week 2 (PBS, NaBu, Resv,). Event counts for all samples before gating was 10.000. Data acquisition and analysis were performed using the free software Floreada.io with Cell Cycle analysis based on the Watson pragmatic model.

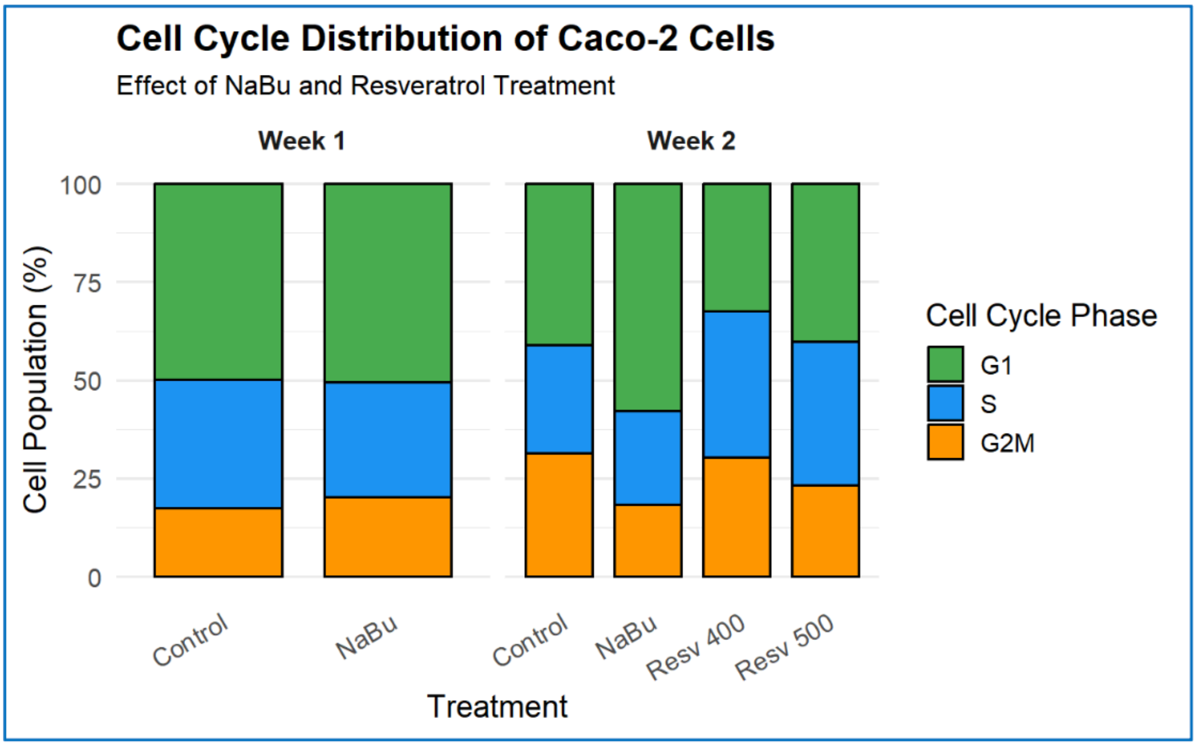


Figure 6. Cell-cycle proportions for the different treatments. For the different weeks. Image taken from Blue's report.

Discussion

Cell cycle analysis estimates proliferation by determining the percentage of cells in G0/G1, S, and G2/M phases. This provides information about the distribution of cells across different phases of the cell cycle, and not just whether the cells are proliferating, which is an advantage because it can identify cell cycle arrest of delays caused by exposure of chemicals, drugs etc. Another advantage of the cell cycle analysis is also that it analyses a large number of cells. Some of the drawback from cell cycle analysis is that it measures DNA content rather than DNA synthesis directly, which makes it an indirect measurement of proliferation, and it cannot distinguish between cells that are actively progressing through the cell cycle and cells that are arrested in a cell cycle phase.

It provides an insight into the cell population at one time (snapshot), so if you want to measure at different timepoint, new experiments should be initiated with different time points for harvesting. Furthermore, cells with abnormal DNA content can complicate the interpretation.

Bromodeoxyuridine (BrdU) incorporation directly measures DNA synthesis because BrdU is incorporated into newly replicated DNA during S phase of the cell cycle. Therefore, it could be interpreted as a more direct measurement of proliferating cells. However, BrdU assays require additional staining procedures, DNA denaturation to expose incorporated BrdU, and may affect

cell physiology. Unlike cell cycle analysis, BrdU labeling specifically identifies cells actively synthesizing DNA.
Cell cycle analysis provides broader information about cell cycle dynamics and checkpoints, whereas BrdU incorporation offers direct measurement of DNA replication and cell proliferation.

The flow cytometry experiments performed in both weeks consistently showed that NaBu altered the DNA-content distribution of DLD-1 cells. Visual inspection of the histograms demonstrated an increased population of cells with DNA content following NaBu treatment, suggesting an accumulation in the G2/M phase of the cell cycle. Although Floreada generated numerical estimates for the G1, S and G2/M fractions, the automatic fitting was suboptimal for the NaBu-treated samples because the fitted S-phase distribution overlapped with the G2/M peak. Consequently, these percentages should be regarded as approximate estimates rather than precise measurements. Our control sample for week 2 shows a distribution that differs a lot from our week 2 experiments by having an enrichment in the proportion of cells in the G2 phase. One possibility could be human error when preparing the treatment for the flask. The sample treated with Resveratrol also shows a very unusual distribution with almost zero cells in G2 phase. However, we must be cautious with overinterpreting here as the Floreada software was not 100% reliable in estimating cell cycle distributions.
The observed accumulation of cells with DNA content is consistent with a NaBu-induced delay or arrest in G2/M and provides a explanation for the reduced proliferation observed in the Hoechst and resazurin assays (see results from Hoechst and resazurin assay). Cells that fail to progress through mitosis divide less frequently, leading to a reduction in cell number while maintaining metabolic activity proportional to the remaining viable cells. Together, the viability assays and flow cytometry therefore support the conclusion that NaBu primarily inhibits proliferation by interfering with cell-cycle progression rather than causing acute cell death.

Hoechst 33342 / Alamar Blue 96 well assays (2b):

Introduction

In this experiment, fluorescence-based viability assays were used to investigate the effects of sodium butyrate (NaBu) on DLD-1 and Caco-2 cell lines. Cell viability and metabolic activity were assessed using the Hoechst 33342 and resazurin (Alamar Blue) assays. Hoechst 33342 binds DNA and provides an estimate of relative cell number, whereas resazurin is converted to a fluorescent product by metabolically active cells and reflects cellular metabolic activity.

Methods

Week 1

DLD-1 and Caco-2 cells were seeded in 96-well plates at 10,000 cells per well. Cells were treated with increasing concentrations of NaBu (0.5, 1, and 5 mM). For viability measurements, cells were incubated with Hoechst 33342 (5 μg/mL) and resazurin (25 μg/mL). Resazurin fluorescence was measured first to estimate metabolic activity. Cells were subsequently washed and Hoechst fluorescence was measured to estimate relative cell number. Results were normalized to untreated control cells.

Week 2

We deviated from the protocol in week 2 by accident and only made one plate with DLD-1 occupying columns 2-6 and Caco-2 occupying columns 7-11. Due to this mistake we could not make a dose response curve, but instead we went with a binary assay using fixed concentrations of 1 mM NaBu and 40 μM Resveratrol, both alone and combined. Cells were incubated with the same Hoechst and resazurin concentrations as in week 1 and fluorescence was measured. Results were normalized to untreated control cells. Note that since columns 2 and 11 in the plate contained cells+media (but not media+vehicle) they were not used for the final analysis (see page the appendix, page 36).

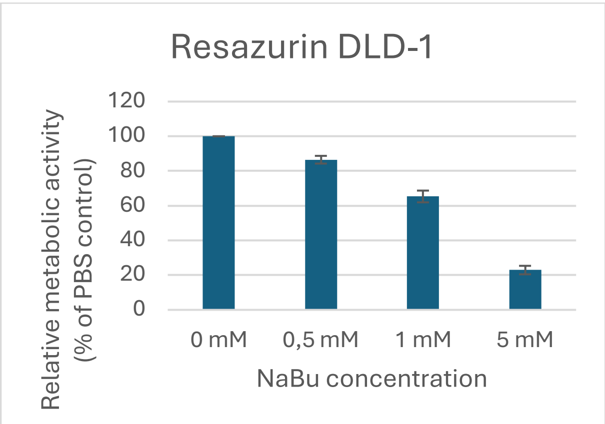
Results

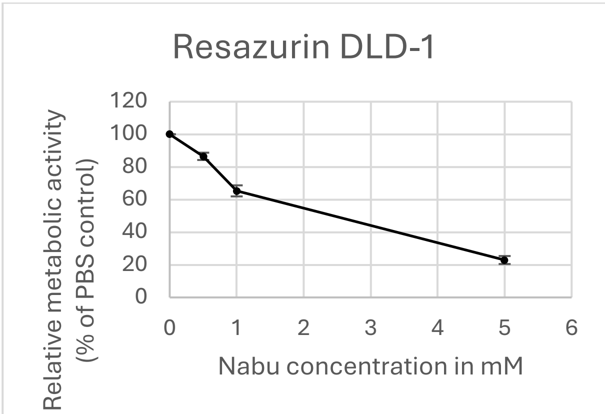
Week 1

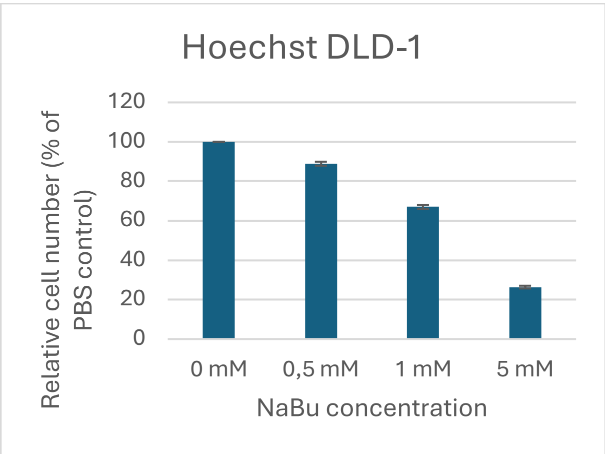
The effect of NaBu on cell number was assessed using the Hoechst 33342 assay, while cellular metabolic activity was evaluated using the resazurin assay.

For DLD-1 cell line NaBu treatment reduced both cell number and metabolic activity in a concentration-dependent manner as seen on figure 5. Relative cell number measured by Hoechst fluorescence decreased from 100% in the control group to approximately 88%, 67%, and 26% following treatment with 0.5, 1, and 5 mM NaBu, respectively.

Similarly, resazurin fluorescence decreased to approximately 87%, 65%, and 23% of control values at 0.5, 1, and 5 mM NaBu, respectively. The reduction in metabolic activity closely followed the reduction in cell number.







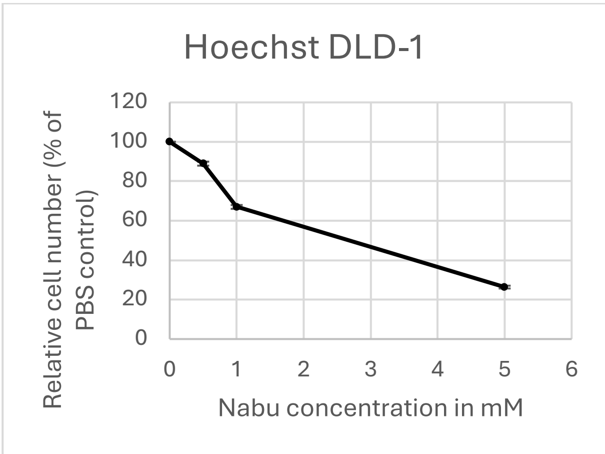


Figure 7. Effect of increasing NaBu concentrations (0, 0.5, 1 and 5 mM) on DLD-1 cells measured using resazurin (metabolic activity) and Hoechst 33342 (relative cell number). Fluorescence values were normalized to untreated controls (100%). Bars represent the mean of replicate measurements, and error bars indicate ± SEM.

Caco-2 cells appeared less sensitive to NaBu treatment than DLD-1 cells. At 0.5 and 1 mM NaBu, Hoechst and resazurin signals remained close to control levels. At 5 mM NaBu, however, cell number decreased to approximately 61% of control and metabolic activity decreased to approximately 57% of control.

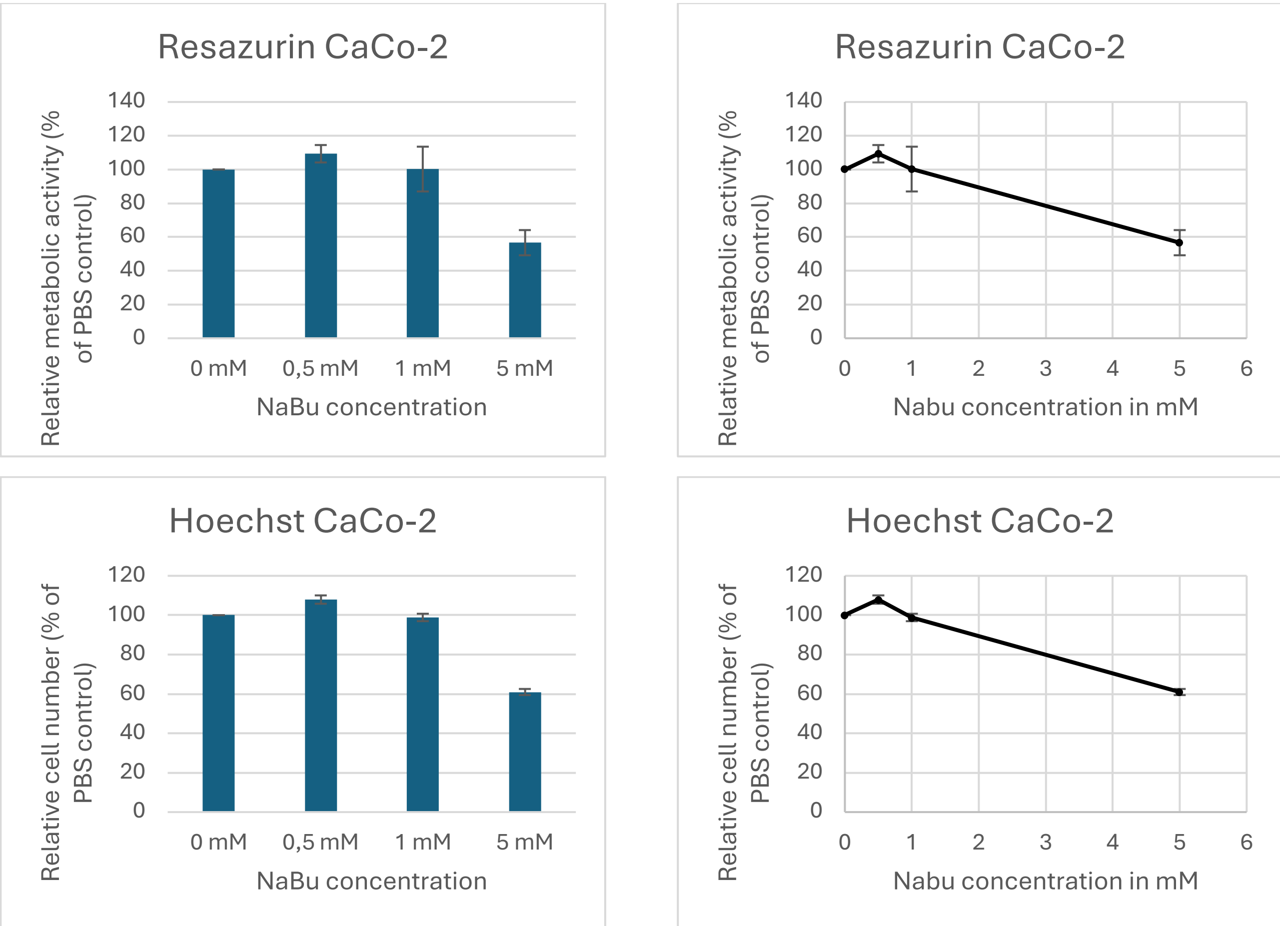


Figure 8. Effect of increasing NaBu concentrations (0, 0.5, 1 and 5 mM) on Caco-2 cells measured using resazurin (metabolic activity) and Hoechst 33342 (relative cell number). Fluorescence values were normalized to untreated controls (100%). Bars represent the mean of replicate measurements, and error bars indicate ± SEM.

Week 2:

The week 2 experiment reinforced the previous NaBu results, with DLD-1 cells responding relatively strongly to 1 mM NaBu by showing a ~30-40% decrease in fluorescence intensity relative to the control for both stains (figure 9). Addition of 40 μM Resveratrol strongly reduces cell proliferation by a factor of ~x10, and the combination of NaBu and Resveratrol shows the same result. For Caco-2 cells, we observe an increase in cell proliferation for both NaBu and Resveratrol treatment, and only the combined treatment shows a level around the control. However, we must note that our Caco-2 samples for week 2 showed a very high variance for multiple treatments, making it harder to draw firm conclusions. In addition, while the relative signal for the two stains is quite aligned for DLD-1 cells it is more decoupled for Caco-2, with the Hoechst signal showing a greater relative effect for all treatments.

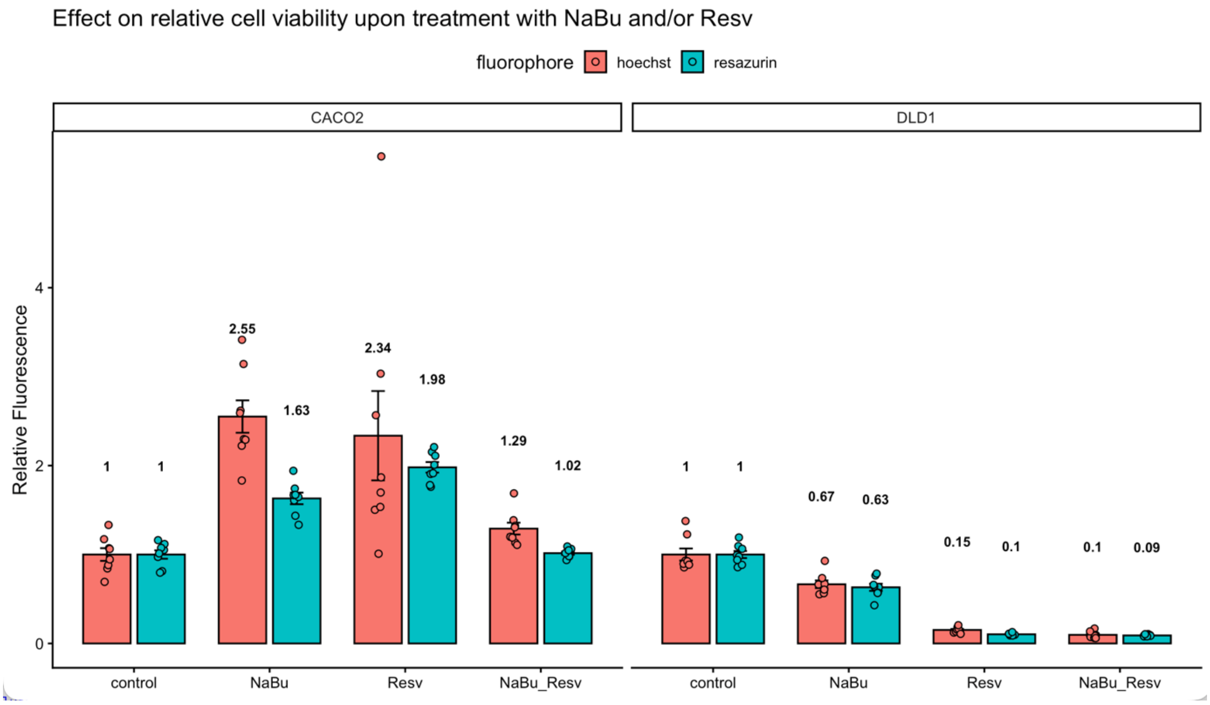


Figure 9, Cell viability measured both by DNA synthesis (Hoechst stain) and cell metabolism (resazurin stain). The control samples used vehicle PBS and DMSO in matching volumes. Concentration of NaBu: 1mM, concentration of Resveratrol: 40 μM. The bar charts indicate the group means relative to the control. Error bars: standard deviation of the mean.

For further comparison, we can look at group Black's data for this experiment, as they did the experiment properly for both cell-lines, and thus get an idea what further data points look like. We see that for DLD-1 the addition of NaBu causes a dose dependent decrease in Hoechst fluorescence. Addition of Resveratrol also causes a decrease, but less pronounced. The combinatorial addition of both compounds then causes an even further decrease (figure 10). The effect is largely mirrored in the resazurin fluorescence, with the exception that at high doses of NaBu further addition of Resveratrol does not cause additional decrease in metabolic activity (figure 11).

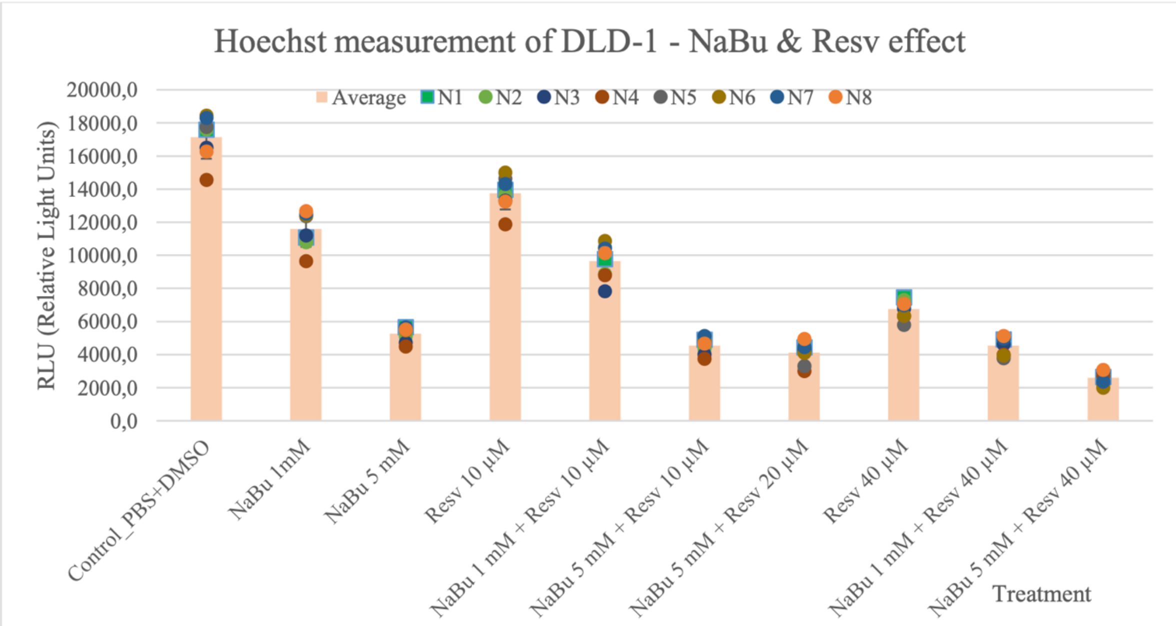


Figure 10. Hoechst fluorescence for the different treatments for DLD-1. Adapted from Black's report.

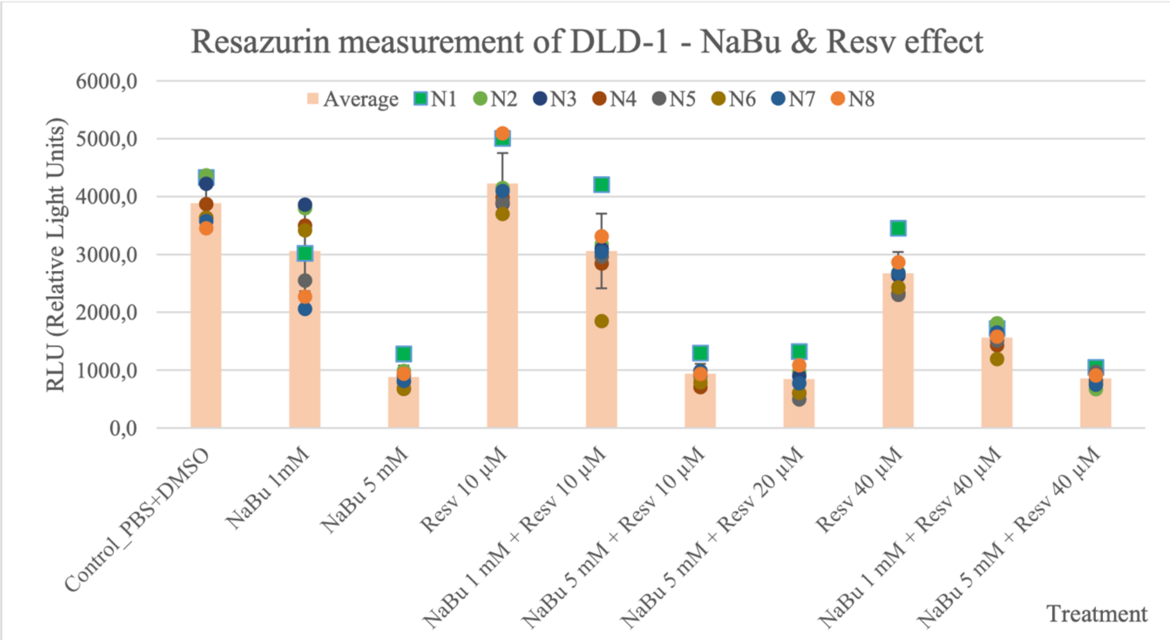


Figure 11: Hoechst fluorescence for the different treatments for DLD-1. Adapted from Black's report.

Looking at the Hoechst data for Caco-2 we see a a similar trend as for DLD-1, which contradicts our own data where Caco-2 cells are not affected, or indeed show higher proliferation in the NaBu/Resv treatments (figure 12). The resazurin data shows a similar pattern except that in some treatments the relative signal also increases. This increased variance highlights that Caco-2 cells look to be more challenging to work with in general (figure 13).

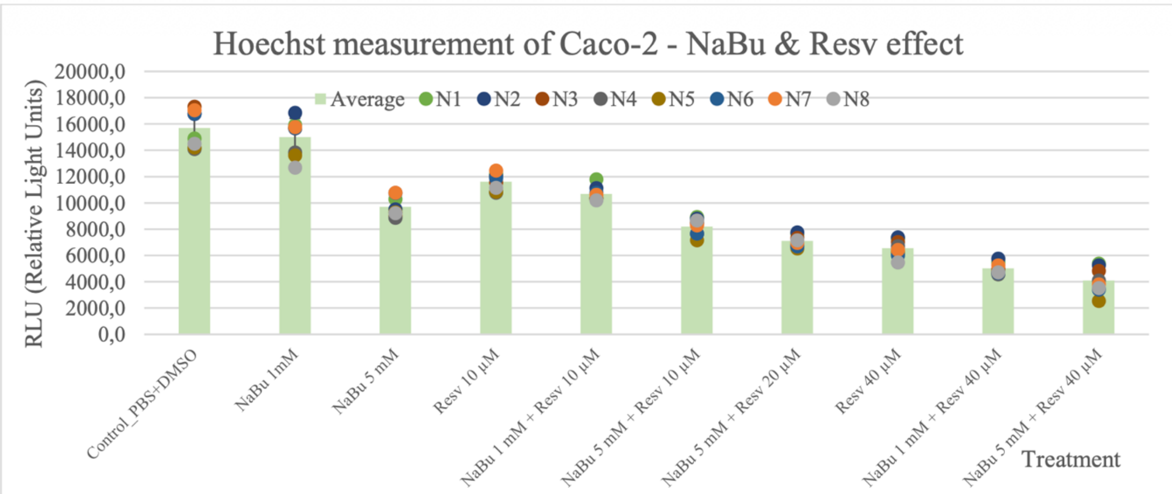


Figure 12: Hoechst fluorescence for the different treatments for Caco-2. Adapted from Black's report.

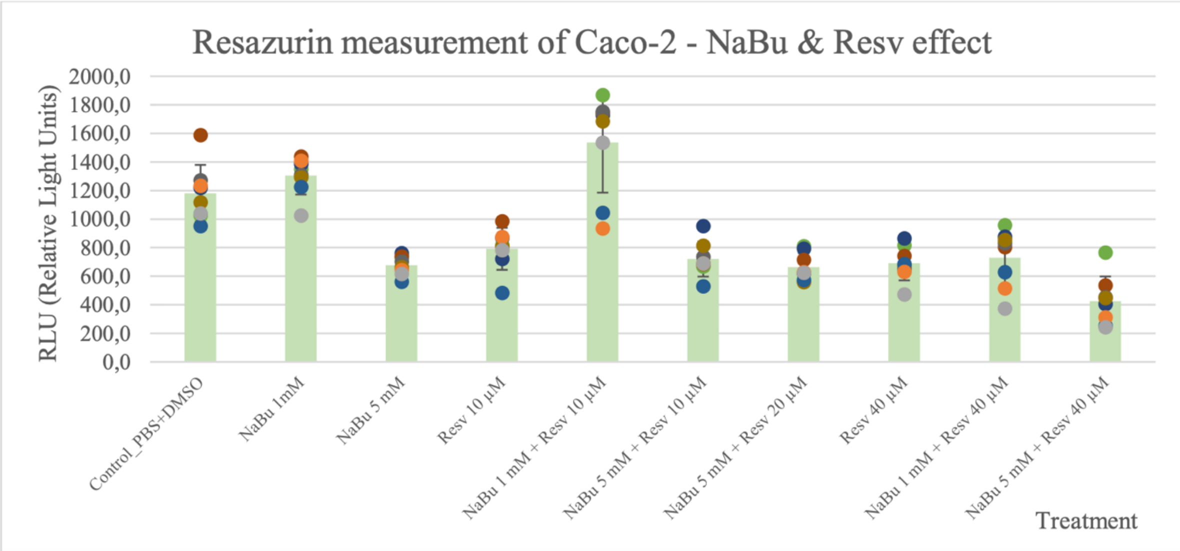


Figure 13. Resazurin fluorescence for the different treatments for Caco-2. Adapted from Black's report.

Discussion

In the week 1 assay, the Hoechst 33342 stain demonstrated that sodium butyrate reduced the relative number of viable cells in both cell lines. However, the effect was substantially greater in DLD-1 cells than in Caco-2 cells. In DLD-1 cells, cell number decreased progressively with increasing NaBu concentration, reaching approximately 26% of control at 5 mM NaBu. In contrast, Caco-2 cells were less affected at lower concentrations and retained approximately 61% of control cell number at 5 mM NaBu. The resazurin assay showed a similar concentration-dependent decrease in metabolic activity. The decrease in resazurin fluorescence closely matched the decrease in Hoechst fluorescence.

For the week 2 assay, we again saw more evidence that DLD-1 cells responded to 1mM NaBu by having reduced proliferation, as evidenced by both Hoechst and resazurin fluorescence being decreased in tandem, while Caco-2 cells in fact showed increased proliferation in our assay. DLD-1 cells showed a very strong response to 40 µM Resveratrol and Resveratrol+NaBu. Due to the uncertainty in our results, especially for the Caco-2 cells, more research needs to be conducted to verify whether this cell line indeed shows a replicable increase in cell proliferation in response to NaBu or Resveratrol.

The flow cytometry results indicated that the NaBu treatment caused an accumulation in the proportion of cells in the G2 phase of the cell cycle and a corresponding decrease in the S-phase population. This indicates that NaBu induced cell cycle arrest, which likely contributed to the reduction in cell proliferation observed in the viability assays.

Conclusion

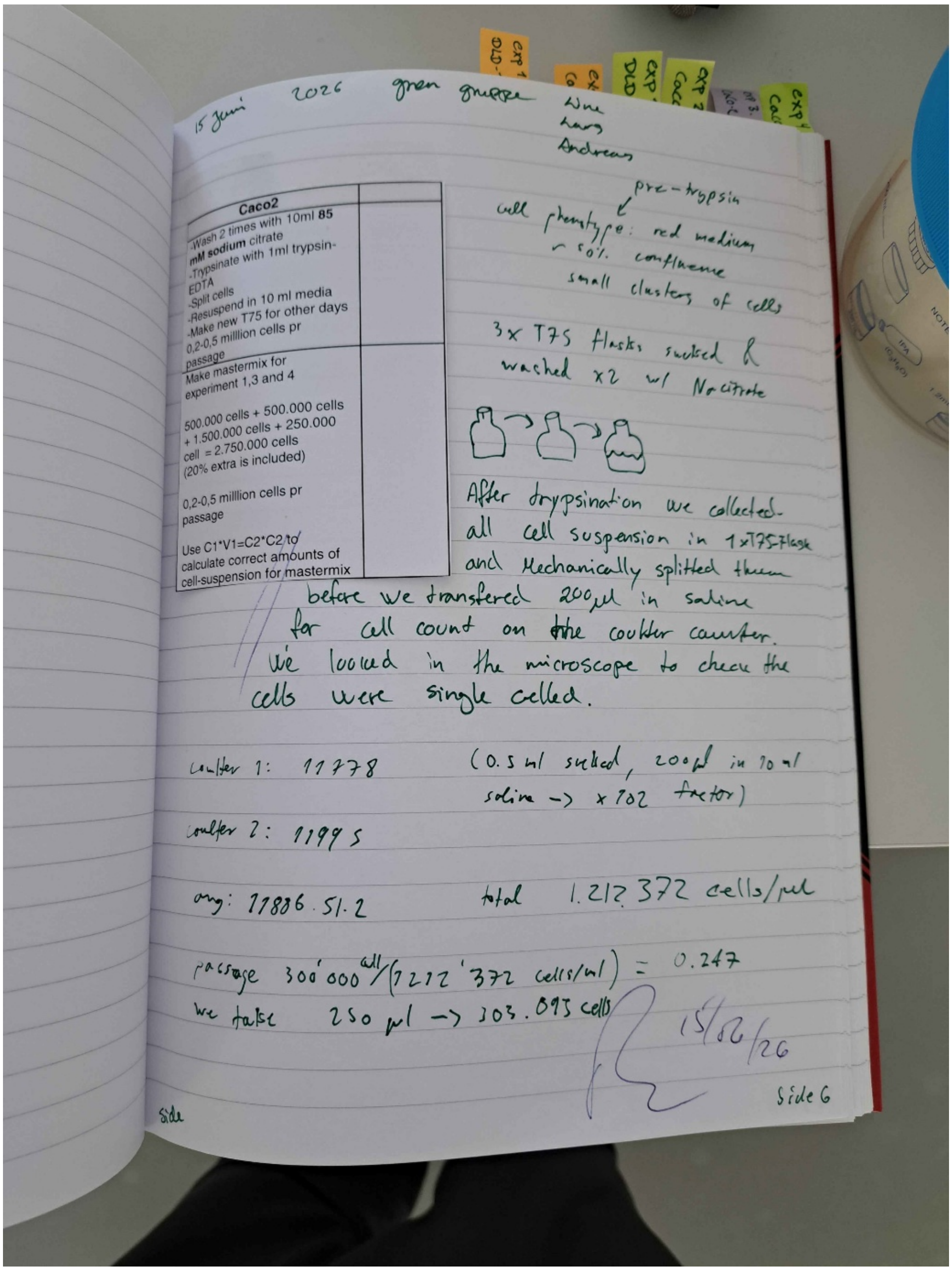
Treatment with NaBu reduced both the relative cell number and metabolic activity of DLD-1 cells in a concentration-dependent manner, whereas Caco-2 cells were less sensitive to the treatment. Resveratrol causes a strong reduction in the overall relative cell number and metabolic activity in DLD-1 cells, while leaving Caco-2 cells more or less unaffected, although the uncercainty for the Caco-2 results warrant further investigation.

Flow cytometry demonstrated that NaBu altered the DNA-content distribution of DLD-1 cells, consistent with an accumulation of DNA content and therefore indicative of cell-cycle arrest. Because the automatic cell-cycle fitting performed by Floreada.io inadequately histograms, the calculated phase percentages should be interpreted cautiously. Nevertheless, the qualitative changes observed in the DNA-content histograms together with the viability assays support the conclusion that NaBu suppresses DLD-1 cell proliferation.

Appendix:

Lab journal:

2A:



15 Juni 2026 green gruppe
Wu
Hans
Andreas

Caco2	
-Wash 2 times with 10ml **85 mM sodium** citrate -Trypsinate with 1ml trypsin-EDTA -Split cells -Resuspend in 10 ml media -Make new T75 for other days 0,2-0,5 milllion cells pr passage	
Make mastermix for experiment 1,3 and 4 500.000 cells + 500.000 cells + 1.500.000 cells + 250.000 cell = 2.750.000 cells (20% extra is included) 0,2-0,5 milllion cells pr passage Use C1*V1=C2*C2 to calculate correct amounts of cell-suspension for mastermix	

cell phenotype: red medium
pre-trypsin
~ 50% confluence
small clusters of cells

3x T75 flasks sucked & washed x2 w/ Na citrate

After trypsination we collected all cell suspension in 1xT75-Flask and Mechanically splitted them before we transfered 200µl in saline for cell count on the coulter counter. We looked in the microscope to check the cells were single celled.

coulter 1: 11778 (0.5 ml sucked, 200µl in 10 ml saline -> x 102 factor)

coulter 2: 11995

avg: 11886.51.2 total 1.212.372 cells/ml

passage 300'000 cell/(1212'372 cells/ml) = 0.247
we take 250 µl -> 303.093 cells

15/06/26

Side Side 6

exp. 2A: DLD-1
- we take two bottles and put 11,21 mL Mcoy in each as well as 0,79 mL DLD-1, ~~with~~ so we can have 200.000 cells in each

exp. 2B: DLD-1
- protocol says to make a master mix of 10 mL, in order to use the multipipet so we will double it to 20mL by doubling 9,44×2 = 18.88 mL mcoy and 0,56×2 = 1,12 mL

- we ~~take~~ put µL of the master mix per well and add 180 µL mcoy per well and put the well plates in heat

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17.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2a:
Cell cycle analysis

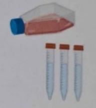
Preparation of cell samples for Flow cytomerty.

- Trypsinize cells. Resuspend cells in PBS-EB (PBS w/o + 5 mM EDTA + 0.5% BSA) and pellet cells by centrifugation for 5 min at appr. 300g, about 1800 rpm. (Note: ~~If you want to analyze apoptotic cells as well, remember to collect the cells floating in the media – but we do not include these cells for this experiment~~). Discard supernatant.
- Wash cells in appr. 10 mL PBS-EB
- Spin down as before. Discard supernatant. Resuspend cells in 3-5 mL PBS-EB.
- Count the cells. Transfer 500.000 cells to a FACS tube (BD, Falcon 5 mL polystyrene round-bottom tube, cat # 35 2058). Add carefully PBS-EB until the total volume is appr. 3 mL. Spin down the cells and discard supernatant. Make up to 3 of these FACS tubes for each cell line.

500.000
800.000

T75-PBS

T75-NaBu



Resuspend cells in exactly 200 μL PBS-EB. Add slowly 2 mL ice-cold 70% ethanol while whirly mixing (IMPORTANT). Incubate on ice for at least 30 min. ~~The samples should be kept at -20°C at this stage.~~

kl. 13:00

wear gloves for

Staining the cells for Flow cytometry

- Centrifuge the cells at 300g and ~~4°C~~ for 5 min. Discard supernatant.
- Resuspend cells in 2-~~3~~ mL PBSw. Spin as before. Discard supernatant.
- Resuspend cells in 400 μL PBSw. Add 50 μL RNAse (1 mg/ mL) and 50 μL propidium iodide (0.4 mg/ mL). Incubate at RT for 30 min in the dark. 15.05
- Centrifuge the cells at 300g and ~~4°C~~ for 5 min. Discard supernatant.
- Resuspend cells in 0,5 mL PBSw
- Analyze by flow cytometry.

different centrifuge (that doesn't cool)

- Washed with PBS% first. 10 mL
- ~~We take our cells from incubation and trypsinize~~ them. ~~We accidently washed the NaBu samples.~~

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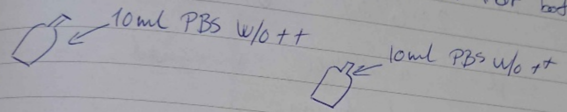
taken out samples from the 37°c incobator,
sucked the media from the T75 Flask
and added 1ml Trypsin in both, Next
we incobated the cells with Trypsin in
37°C for 6min.
we looked in the microscope to confirme
de-attachment and then resuspended the
cells in 13ml PBS W/o + EDTA + BSA for both
T75-Flasks



the Flask without NaBU didnt De-attached
from the Flask so we washed them
with PBS % another time to make
sure and then try to trypzinate again

- we remove liquid after centrifugation
until we have 0,5 mL and pellet at the bottom
of the flask to centrifuge again.

- we resuspend the pellet in PBS w/o + EDTA

- We centrifuged at 1800 RPM and
prepared the cells for cell counting ; coulter
Obs: first centrifugation was at 1264 rpm

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17-06-2026

- We labled 4 plastic cups, two for each for a beyyer result. 200 µL in each

PBS: 1: 5281 — 4876
2: 4649 — 4640 — avg 4976

NaBu: 1: 6261 — 6765
2: 6909 — 6750 — avg. 6573

200 µl in 16 ml NaCl, 0.5 ml sucked by machine
-> x 762 fold dilution

PBS: 497'148 cells/ml
NaBu: 670'242 cells/ml

need 500'000 cells for FACS ->
PBS: ~~1 ml of cells in each tube~~ 0.5 ml/497148
NaBu: 500'000 cells / 670242 cells/ml = 0.746

~~B.B.~~

~~0,746 × 4 = 2,984~~ ~~0,6~~

We have 3 FACS-Tubes and will use ~~that~~ ~~in each tube~~. Use 1,006 ml ~~for nabu~~ = for the PBS and 0,746 ml the Nabu

FS

18.06.2026 green team: Lin
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blu
Perform viability assay:
Prepare growing medium:
6 ml DMEM + 3 ul Ho
Resazurin
6 ml McCoy + 3 ul Ho
Resazurin
Remove media from 96-well p
100 ul of prepared media in e
incubate for 30 min at 37C in
Read the resazurin fluorescen
immediately:
a. Synergy HT, Bi
reader (28A.2-
b. Excitation 530/
wheel D) and
c. Emission 590/2
wheel A), respe
45
Wash three times with 100µL P
Add 100µL PBSw as the last ste
Read the Hoechst fluorescence
d. Synergy HT BioTe
reader (28A.2-21)
e. Excitation 360/40
wheel D)
f. Emission 460/40
wheel A), Gain: 60

18.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin
- 6 ml McCoy + 3 ul Hoechst + 15 ul Resazurin

Remove media from 96-well plate and add 100 ul of prepared media in each well. - incubate for 30 min at 37C in incubator.

Read the resazurin fluorescence immediately:

a. Synergy HT, BIOTEK plate reader (28A.2-21)
b. Excitation 530/25 nm (Filter wheel D) and
c. Emission 590/20 nm (Filter wheel A), respectively. Gain: 45

Wash three times with 100µL PBSw.

Add 100µL PBSw as the last step.

Read the Hoechst fluorescence

d. Synergy HT BioTek plate reader (28A.2-21)
e. Excitation 360/40 nm (Filter wheel D)
f. Emission 460/40 nm (Filter wheel A), Gain: 60.

19.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2a:
Cell cycle analysis: Analyse FACS data with Floreada/Flowing software

19.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays
Do statistics on effect of NaBu on Hoechst/resaruzin.

28

22.06.2026 green team: Line, Lars, Andreas
DLD-1:
Seed for experimental 2A: cell cycle analysis

Seed 3 x T75 flask with 700.000 cells in each flask, total of 12 ml/flask.

1153 and 10649 cells pr 0,5 ml from coulter counter

$$\frac{1153 + 10649}{2} = 11.090 \cdot 51 \cdot 2 = 1.131.180 \text{ cells/ml}$$

$$\frac{700.000}{1.131.180} = \underbrace{0,618 \text{ ml}}_{\text{cell suspension}} + 12 \text{ ml media}$$

22.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays

	1	2	3	4	5	6	7	8	9	10	11	12
A–H	No cells – media only	Control: PBS+DMSO	1 mM NaBu + DMSO	5 mM NaBu + DMSO	PBS + 10 µM Resv	1 mM NaBu + 10 µM Resv	5 mM NaBu + 10 µM Resv	5 mM NaBu + 20 µM Resv	PBS + 40 µM Resv	1 mM NaBu + 40 µM Resv	5 mM NaBu + 40 µM Resv	No cells – media only

Figure 2.2

Seed one 96 well plate with 10.000 DLD-1 cells pr well. Volume of 200 uL pr well.

Mastermix:
10ml with 500.000 cells

$$\frac{500.000}{1.131.180} = 0,442 \text{ cell sup} + 9,9558 \text{ media}$$

22/06/24

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23.06.2026 green team: Line, Lars, Andreas
DLD-1
Experiment 2a: cell cycle
Treat with PBSw, 20 uM Resv and 10 mM NaBu

23-06-2026 green team: Line, Lars, Andreas
DLD-1: McCoy medium
Experiment 2A:
Need PBSw, 20 µM Resv and ~~5~~ 10 mM NaBu
75 µl PBS + 3µl DMSO in 15 ml media
75 µl PBS + 3 µl Resveratrol in 15 ml media
Use 75 µl NaBu 3 µl DMSO in 15 ml media

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24.06.2026 green team: Line, Lars, Andreas
DLD-1
Experiment 2a: cell cycle
FACS
Input: 3x treated T75 flasks
Workflow:

- Trypsinize
- Resuspend in 9 ml PBS-EB
- Centrifugate for 5 min at 300g then discard supernatant
- Wash in 10 ml PBS-EB
- Centrifugate 5 min at 300g then n discard supernatant — not enough, 40 with 9 mL
- Resuspend in 3 ml PBS-EB
- Count cells
- Transfer 500'000 cells to FACS tube fill until 3 ml. → with PBS-EB
- Generate more tubes if enuff cells
- Centrifugate and discard supernatant
- Resuspend in 200 µl PBS-EB
- Add 2 ml ice-cold 70% ethanol while whirly mixing (vortex). ← slowly
- Incubate on ice for at least 30 minutes. Keep samples on -20.
- ---- staining step ----------------
- Centrifugate at 300g for 5 min. Discard supernatant.
- Resuspend in 2-3ml PBSw.
- Centrifugate at 300g for 5 min. Discard supernatant
- Resuspend in 400 µl PBSw. Add 50 µl RNAse and 50 µl propidium iodide (use gloves). Incubate at room temperature for 30 min.
- Centrifugate at 300g for 5 min. Discard supernatant.
- Resuspend in 0.5 ml PBSw.
- Analyze by flow cytometry

confluenes:
Ctrl: 40%
Resv: 90%
NaBu: 33%

			cells/µl concentration
Counts:	ctrl 1:	673	62525
200 µl	ctrl 2:	698	71196
x102	NaBu1:	793	80886
	NaB2:	891	90882
	Resv1:	447	45594
	Resv2:	535	54570

volumes after (looking at 15 ml tube)
Ctrl: 2.5 ml
Resv: 2.5 ml
NaBu: 2.6 ml

FACS seems fine (number of cells)

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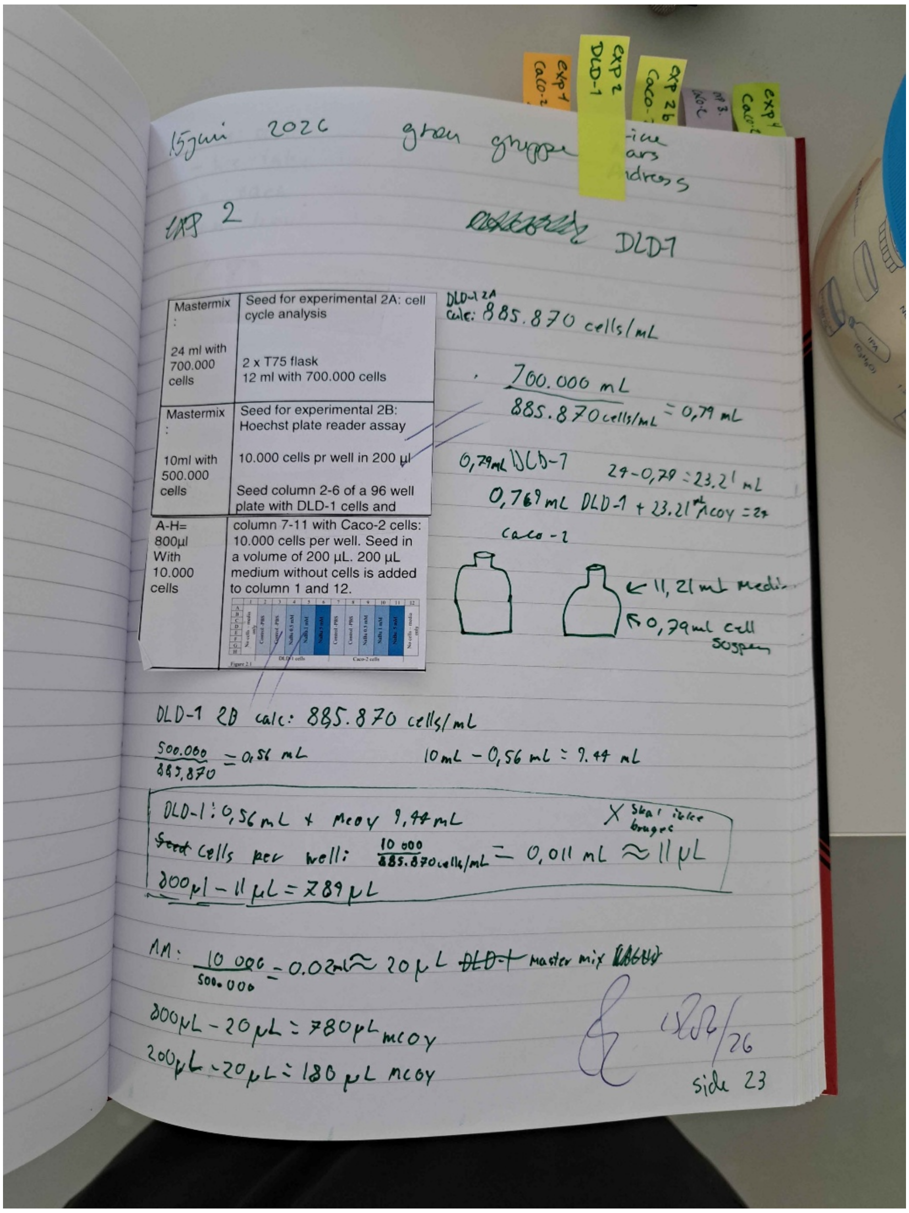
Experiment 2B:



15 juni 2026 grøn gruppe

EXP 2 DLD-1

Mastermix: 24 ml with 700.000 cells	Seed for experimental 2A: cell cycle analysis 2 x T75 flask 12 ml with 700.000 cells
Mastermix: 10ml with 500.000 cells	Seed for experimental 2B: Hoechst plate reader assay 10.000 cells pr well in 200 µl Seed column 2-6 of a 96 well plate with DLD-1 cells and
A-H= 800µl With 10.000 cells	column 7-11 with Caco-2 cells: 10.000 cells per well. Seed in a volume of 200 µL. 200 µL medium without cells is added to column 1 and 12.

Figure 2.1

DLD-1 2A calc: 885.870 cells/mL

$\frac{700.000 \text{ mL}}{885.870 \text{ cells/mL}} = 0,79 \text{ mL}$

0,79mL DLD-1 24-0,79 = 23,21 mL

0,769 mL DLD-1 + 23.21 mL mcoy = 24

Caco-2

← 11,21 mL medi.

← 0,79mL cell suspen

DLD-1 2B calc: 885.870 cells/mL

$\frac{500.000}{885.870} = 0,56 \text{ mL}$ 10mL − 0,56 mL = 9.44 mL

DLD-1: 0,56mL + mcoy 9,44mL X Skal ikke bruges

~~Seed~~ Cells per well: $\frac{10\,000}{885.870 \text{ cells/mL}} = 0,011 \text{ mL} \approx 11 \mu L$

200µl − 11µL = 789µL

MM: $\frac{10\,000}{500.000} = 0.02\text{ml} \approx 20 \mu L$ ~~DLD-1~~ master mix

800µL − 20µL = 780µL mcoy

200µL − 20µL = 180µL mcoy

15/06/26

side 23

exp. 2A: DLD-1

- we take two bottles and put 11,21 mL Mcoy in each as well as 0,77 mL DLD-1, ~~with~~ so we can have 700.000 cells in each

exp. 2B: DLD-1

- protocol says to make a master mix of 10 mL, in order to use the multipipet so we will double it to 20mL by doubling 9,44x2= 18.88 mL mcoy and 0,56x2=1,12 mL

- we ~~take~~ put µL of the master mix per well and add 180 µL mcoy per well and put the well plates in heat

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16.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2a: Cell cycle analysis

NaBu-treated flask (10 mM final)

- 120 µL of 1 M NaBu stock
- 11,88 mL medium

Total = 12 mL

PBS vehicle control flask

To keep the PBS concentration identical to the treated flask:

- 120 µL PBSw
- 11,88 mL medium

Total = 12 mL

16.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b: Hoechst 33342 / Alamar Blue
96 well assays

Prepare 2 mL of each condition

Final NaBu	100 mM NaBu stock	Additional PBSw	Medium	Total
~~Control (0 mM)~~	~~0 µL~~	~~100 µL~~	~~1900 µL~~	~~2000 µL~~
0.5 mM	10 µL	90 µL	1900 µL	2000 µL
1 mM	20 µL	80 µL	1900 µL	2000 µL
5 mM	100 µL	0 µL	1900 µL	2000 µL

Rows	1	2	3	4	5	6	7	8	9	10	11	12
A–H	No cells – media only	Control -PBS	Control -PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	Control -PBS	Control -PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	No cells – media only
		DLD-1 cells					Caco-2 cells					

Figure 2.1

Obs: The wells show different colours depending on either DLD-1 or CaCo-2

- Old medium is removed and medium with NaBu is added. 2 and 3 with PBS control, ~~and~~ 4 with 0,5 mM NaBu, 5 with 1 mM NaBu, 6 with 5 mM NaBu

26/06/26

18.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin
- 6 ml McCoy + 3 ul Hoechst + 15 ul Resazurin

Remove media from 96-well plate and add 100 ul of prepared media in each well. - incubate for 30 min at 37C in incubator.

Read the resazurin fluorescence immediately:

a. Synergy HT, BIOTEK plate reader (28A.2-21)
b. Excitation 530/25 nm (Filter wheel D) and
c. Emission 590/20 nm (Filter wheel A), respectively. Gain: 45

Wash three times with 100µL PBSw.

Add 100µL PBSw as the last step.

Read the Hoechst fluorescence

d. Synergy HT BioTek plate reader (28A.2-21)
e. Excitation 360/40 nm (Filter wheel D)
f. Emission 460/40 nm (Filter wheel A), Gain: 60.

19.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2a:
Cell cycle analysis: Analyse FACS data with Floreada/Flowing software

19.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays
Do statistics on effect of NaBu on Hoechst/resaruzin.

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16.06.2026 green team: Line, Lars, Andreas
CaCo-2
Experiment 2b: Hoechst 33342 / Alamar Blue 96 well assays

Prepare 2 mL of each condition

Final NaBu	100 mM NaBu stock	Additional PBSw	Medium	Total
Control (0 mM)	0 µL	~~100 µL~~	~~1900 µL~~	~~2000 µL~~
0.5 mM	10 µL	90 µL	1900 µL	2000 µL
1 mM	20 µL	80 µL	1900 µL	2000 µL
5 mM	100 µL	0 µL	1900 µL	2000 µL

	1	2	3	4	5	6	7	8	9	10	11	12
A–H	No cells – media only	Control -PBS	Control -PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	Control -PBS	Control -PBS	NaBu 0.5 mM	NaBu 1 mM	NaBu 5 mM	No cells – media only
		DLD-1 cells					Caco-2 cells					

Figure 2.1

exp 4
CaCo-2

16/06/26

18.06.2026 green team: Line, Lars, Andreas
CaCo-2:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin

- 6 ml McCoy + 3 ul Hoechst + 15 ul Resazurin

Remove media from 96-well plate and add 100 ul of prepared media in each well. -incubate for 30 min at 37C in incubator.

Read the resazurin fluorescence immediately:

g. Synergy HT, BIOTEK plate reader (28A.2-21)
h. Excitation 530/25 nm (Filter wheel D) and
i. Emission 590/20 nm (Filter wheel A), respectively. Gain: 45

Wash three times with 100µL PBSw.

Add 100µL PBSw as the last step.

Read the Hoechst fluorescence

j. Synergy HT BioTek plate reader (28A.2-21)
k. Excitation 360/40 nm (Filter wheel D)
l. Emission 460/40 nm (Filter wheel A), Gain: 60.

18/06/26

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Exp 1

week 2

Exp 2
DLD1

week 2
Exp 2

22.06.2026 green team: Line, Lars, Andreas
DLD-1:
Seed for experimental 2A: cell cycle analysis

Seed 3 x T75 flask with 700.000 cells in each flask, total of 12 ml/flask.

1153 and 10649 cells pr 0,5 ml from coulter counter

$$\frac{1153 + 10649}{2} = 11.090 \cdot 51 \cdot 2 = 1.131.180 \text{ cells/ml}$$

$$\frac{700.000}{1.131.180} = \underbrace{0,618 \text{ ml}}_{\text{cell suspension}} + 12 \text{ ml media}$$

22.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b:
Hoechst 33342 / Alamar Blue 96 well assays

	1	2	3	4	5	6	7	8	9	10	11	12
A, B, C, D, E, F, G, H	No cells – media only	Control: PBS+DMSO	1 mM NaBu + DMSO	5 mM NaBu + DMSO	PBS + 10 µM Resv	1 mM NaBu + 10 µM Resv	5 mM NaBu + 10 µM Resv	5 mM NaBu + 20 µM Resv	PBS + 40 µM Resv	1 mM NaBu + 40 µM Resv	5 mM NaBu + 40 µM Resv	No cells – media only

Figure 2.2

Seed one 96 well plate with 10.000 DLD-1 cells pr well. Volume of 200 uL pr well.

Mastermix:
10ml with 500.000 cells

$$\frac{500.000}{1.131.180} = 0,442 \text{ cell sup} + 9,4558 \text{ media}$$

22/06/24

66

25/6-2026, Green team, EXP 2B. ~~Colo2~~ and DLD1.

200µl / well. 96-well plate.

10µl 100mM ResV
90µl DMSO → 100µl 10mM ResV.

McCoy
~~DMEM~~

2000µl Media

McCoy
~~DMEM~~
+
PBS + DMSO

1990 Media
2µl PBS
8µl DMSO

1mM Nabu
+
DMSO

1990 media
2µl Nabu
8µl DMSO

40µM ResV
+
PBS

1990 media.
2µl PBS
8µl 40µM ResV 10mM

40µM ResV
+
1mM Nabu

1990µl Media
2µl Nabu
8µl ResV 10mM

20µM ResV
1mM Nabu

1990µl Media
4µl DMSO
2µl Nabu.
4µl ResV. 10mM.

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Exp3 CaCo2

Exp4 CaCo2

22.06.2026 green team: Line, Lars, Andreas

CaCo-2:

Experiment 2b:

Hoechst 33342 / Alamar Blue 96 well assays

	1	2	3	4	5	6	7	8	9	10	11	12
A–H	No cells – media only	Control: PBS+DMSO	1 mM NaBu + DMSO	5 mM NaBu + DMSO	PBS + 10 µM Resv	1 mM NaBu + 10 µM Resv	5 mM NaBu + 10 µM Resv	5 mM NaBu + 20 µM Resv	PBS + 40 µM Resv	1 mM NaBu + 40 µM Resv	5 mM NaBu + 40 µM Resv	No cells – media only

Figure 2.2

Seed one 96 well plate with 10.000 Caco-2 cells pr well. Volume of 200 uL

Mastermix:

10ml with 500.000 cells

$$\frac{500\,000 \text{ cells}}{195.075 \text{ cells/mL}}$$

$= 2.56\text{mL}$

$10 - 2.563\text{mL} = 7.436\text{mL}$ media

7–11 filled with CaCo2 cells (500.000 cells)

22/06/2026

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CaCO2

Exp 4 CaCO2

23/6-2026, Green team, EXP 2B. CaCO2

200µl /well. 96-well plate.

10µl 100mM ResV → 100µl 10mM ResV.
90µl DMSO

DMEM — 200µl Media

DMEM + PBS + DMSO — 1990 Media, 2µl PBS, 8µl DMSO

1mM Nabu + DMSO — 1990 media, 2µl Nabu, 8µl DMSO

40µM ResV + PBS — 1990 media, 2µl PBS, 8µl 40µM ResV 10mM

40µM ResV + 1mM Nabu — 1990µl Media, 2µl Nabu, 8µl ResV 10mM

20µM ResV 1mM Nabu — 1990µl Media, 4µl DMSO, 2µl Nabu, 4µl ResV. 10mM.

23/6-2026, Green team, EXP 2B. CaCO2

200µl /well. 96-well plate.

10µl 100mM ResV → 100µl
90µl DMSO → 10mM ResV.

DMEM		200µl Media
DMEM + PBS + DMSO		1990 Media 2µl PBS 8µl DMSO
1mM Nabu + DMSO		1990 media 2µl Nabu 8µl DMSO
40µM ResV + PBS		1990 media 2µl PBS 8µl 40µM ResV 10
40µM ResV + 1mM Nabu		1990µl Media 2µl Nabu 8µl ResV 10mM
20µM ResV 1mM Nabu		1990µl Media 4µl DMSO 2µl Nabu. 4µl ResV. 10mM.

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25.06.2026 green team: Line, Lars, Andreas
DLD-1:
Experiment 2b: As described for week 1
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin
- 6 ml McCoy + 3 ul Hoechst + 15 ul Resazurin

Remove media from 96-well plate and add 100 ul of prepared media in each well. - incubate for 30 min at 37C in incubator.

Read the resazurin fluorescence immediately:

a. Synergy HT, BIOTEK plate reader (28A.2-21)
b. Excitation 530/25 nm (Filter wheel D) and
c. Emission 590/20 nm (Filter wheel A), respectively. Gain: 45

Wash three times with 100μL PBSw.

Add 100μL PBSw as the last step.

Read the Hoechst fluorescence

d. Synergy HT BioTek plate reader (28A.2-21)
e. Excitation 360/40 nm (Filter wheel D)
f. Emission 460/40 nm (Filter wheel A), Gain: 60.

~~Cell check~~
~~Ctrl 1~~:
~~Ctrl 2~~:
~~CRISPR 1~~:
~~CRISPR 2~~:
~~MUCK 1~~:
~~MUCK 2~~:

25-06-2026

25.06.2026 green team: Line, Lars, Andreas
CaCo-2:
Experiment 2b: As described for week 1
Hoechst 33342 / Alamar Blue 96 well assays

Perform viability assay:

Prepare growing medium:

- 6 ml DMEM + 3 ul Hoechst + 15 ul Resazurin
- 6 ml McCoy + 3 ul Hoechst + 15 ul Resazurin

Remove media from 96-well plate and add 100 ul of prepared media in each well. -incubate for 30 min at 37C in incubator.

Read the resazurin fluorescence immediately:

g. Synergy HT, BIOTEK plate reader (28A.2-21)
h. Excitation 530/25 nm (Filter wheel D) and
i. Emission 590/20 nm (Filter wheel A), respectively. Gain: 45

Wash three times with 100µL PBSw.

Add 100µL PBSw as the last step.

Read the Hoechst fluorescence

j. Synergy HT BioTek plate reader (28A.2-21)
k. Excitation 360/40 nm (Filter wheel D)
l. Emission 460/40 nm (Filter wheel A), Gain: 60.

83